

Problem 1

Let $\mathcal{F}$ and $\mathcal{G}$ be σ -fields of subsets of Ω .

- (a) Use elementary set operations to show that $\mathcal{F}$ is closed under countable intersections; that is, if $A_1, A_2, \dots$ are in $\mathcal{F}$, then so is $\bigcap_i A_i$.
- (b) Let $\mathcal{H} = \mathcal{F} \cap \mathcal{G}$ be the collection of subsets of Ω lying in both $\mathcal{F}$ and $\mathcal{G}$. Show that $\mathcal{H}$ is a σ -field.
- (c) Show that $\mathcal{F} \cup \mathcal{G}$, the collection of subsets of Ω lying in either $\mathcal{F}$ or $\mathcal{G}$, is not necessarily a σ -field.

Proof.

(a) :Let $A_1, A_2, \dots \in \mathcal{F}$. Then $A_1^c, A_2^c, \dots \in \mathcal{F}$.

Since $\mathcal{F}$ is a σ -field, we have $\bigcup_i A_i \in \mathcal{F}$.

By De Morgan's Law, $\left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c \in \mathcal{F}$.

Hence, $\bigcap_i A_i \in \mathcal{F}$.

(b) :1. Since $\Omega \in \mathcal{F}$ and $\Omega \in \mathcal{G}$, we have $\Omega \in \mathcal{H}$.

2. $\forall A \in \mathcal{H}$, we have $A \in \mathcal{F}$ and $A \in \mathcal{G}$, so $A^c \in \mathcal{F}$ and $A^c \in \mathcal{G}$, hence $A^c \in \mathcal{H}$.

3. $\forall A_1, A_2, \dots \in \mathcal{H}$, we have $A_1, A_2, \dots \in \mathcal{F}$ and $A_1, A_2, \dots \in \mathcal{G}$,

so $\bigcup_i A_i \in \mathcal{F}$ and $\bigcup_i A_i \in \mathcal{G}$, hence $\bigcup_i A_i \in \mathcal{H}$.

(c) :Counterexample: Let $\Omega = \{1, 2, 3\}$, $\mathcal{F} = \{\emptyset, \{1, 2\}, \{3\}, \Omega\}$, $\mathcal{G} = \{\emptyset, \{1\}, \{2, 3\}, \Omega\}$.

Then $\{1, 2\} \in \mathcal{F}$ and $\{1\} \in \mathcal{G}$, but $\{1, 2\} \cup \{1\} = \{1, 2\} \notin \mathcal{F} \cup \mathcal{G}$.

Hence, $\mathcal{F} \cup \mathcal{G}$ is not necessarily a σ -field.

□

Problem 2

Let $A_1, A_2, \dots$ be a sequence of events. Define

$$B_n = \bigcup_{m=n}^{\infty} A_m, \quad C_n = \bigcap_{m=n}^{\infty} A_m$$

Clearly $C_n \subseteq A_n \subseteq B_n$. The sequences $\{B_n\}$ and $\{C_n\}$ are decreasing and increasing respectively with limits

$$\lim B_n = B = \bigcap_n B_n = \bigcap_n \bigcup_{m \geq n} A_m, \quad \lim C_n = C = \bigcup_n C_n = \bigcup_n \bigcap_{m \geq n} A_m$$

The events B and C are denoted $\limsup_{n \rightarrow \infty} A_n$ and $\liminf_{n \rightarrow \infty} A_n$ respectively. Show that

- (a) $B = \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many values of } n\}$,
- (b) $C = \{\omega \in \Omega : \omega \in A_n \text{ for all but finitely many values of } n\}$,

We say that the sequence $\{A_n\}$ converges to $A = \lim A_n$ if B and C are the same set A .

Suppose that $A_n \rightarrow A$ and show that

- (c) A is an event, i.e. $A \in \mathcal{F}$,
- (d) $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$.

Proof.

$$(a) : \text{Since by definition: } B = \bigcap_n \bigcup_{m \geq n} A_m$$

$$\Rightarrow \text{Suppose } \omega \in B, \text{ then } \omega \in \bigcup_{m \geq n} A_m \text{ for all } n \in \mathbb{N}$$

This means for each n , we can find an $m \geq n$ such that $\omega \in A_m$

Hence, $\omega \in A_n$ for infinitely many values of n

$$\Leftarrow \text{Suppose } \omega \in A_n \text{ for infinitely many values of } n,$$

then for each n , we can find an $m \geq n$ such that $\omega \in A_m$

$$\text{This means } \omega \in \bigcup_{m \geq n} A_m \text{ for all } n \in \mathbb{N}$$

$$\text{Hence, } \omega \in \bigcap_n \bigcup_{m \geq n} A_m = B$$

$$(b) : \Rightarrow \text{Suppose } \omega \in C, \text{ then } \exists n \in \mathbb{N} \text{ such that } \omega \in \bigcap_{m \geq n} A_m$$

This means $\exists n \in \mathbb{N}$ such that $\omega \in A_m$ for all $m \geq n$

Hence, $\omega \in A_n$ for all but finitely many values of n

$$\Leftarrow \text{Suppose } \omega \in A_n \text{ for all but finitely many values of } n,$$

then $\exists n \in \mathbb{N}$ such that $\omega \in A_m$ for all $m \geq n$

$$\text{This means } \omega \in \bigcap_{m \geq n} A_m$$

$$\text{Hence, } \omega \in \bigcup_n \bigcap_{m \geq n} A_m = C$$

(c) : Since $A_n \rightarrow A$, so $\lim_{n \rightarrow +\infty} A_n = \limsup_{n \rightarrow +\infty} A_n = \liminf_{n \rightarrow +\infty} A_n$

So we have $A = B = C$

Since B_n are countable unions of A_m , so $B_n \in \mathcal{F}$ for all $n \in \mathbb{N}$

Then, $B = \bigcap_n B_n \in \mathcal{F}$

Hence, we have $A \in \mathcal{F}$

(d) : Since $C_n \subseteq A_n \subseteq B_n$, we have $\mathbb{P}(C_n) \leq \mathbb{P}(A_n) \leq \mathbb{P}(B_n)$

Since C_n increases to C , B_n decreases to B ,

we have $\mathbb{P}(C_n) \rightarrow \mathbb{P}(C) = \mathbb{P}(A)$, $\mathbb{P}(B_n) \rightarrow \mathbb{P}(B) = \mathbb{P}(A)$

Hence by squeeze theorem, $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$

□

Problem 3

Let $\mathcal{E}$ be the left open right closed intervals of $\Omega := \mathbb{R}$ defined in class. Write down the algebra generated by $\mathcal{E}$, and prove your result.

We have $\mathcal{E} = \{\text{left open right closed intervals}\} = \begin{cases} (a, b], & -\infty \leq a < b < +\infty \\ (a, +\infty) \end{cases}$

$$\begin{aligned} a(\mathcal{E}) &= \text{"finite disjoint union of elements in } \mathcal{E}\text{"} \\ &= \underbrace{\{I_1 \cup \dots \cup I_k; I_j \in \mathcal{E}, \text{ and } I_i \cap I_j = \emptyset\}}_f \end{aligned}$$

Proof. Denote the collection of finite disjoint union of elements in $\mathcal{E}$ by f

- $f \subseteq a(\mathcal{E})$
 - Since $a(\mathcal{E})$ is an algebra containing $\mathcal{E}$, it is closed under finite unions.
 - Any element of f is a finite disjoint union of intervals from $\mathcal{E}$, hence belongs to $a(\mathcal{E})$.
 - Thus, $f \subseteq a(\mathcal{E})$
- $a(\mathcal{E}) \subseteq f$
 - we need to prove that f is an algebra
 - * since $\Omega = \mathbb{R} = (-\infty, +\infty)$, choose $I_1 = (-\infty, a], I_2 = (a, +\infty) \in \mathcal{E}$, then $I_1 \cup I_2 = \Omega \in f$
 - * for $\forall I_i, I_j \in f$, $I_i \cup I_j \in f$ by definition

- * for $\forall(a, b] \in f$, we have $(a, b]^c = (-\infty, a] \cup (b, +\infty) \in f$
- for $\forall(a, +\infty) \in f$, we have $(a, +\infty)^c = (-\infty, a] \in f$
- since $\mathcal{E} \subseteq f$, f is an algebra and $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$,
Hence, $a(\mathcal{E}) \subseteq f$

□

Problem 4

Let Ω be a sample space. Show that any finite algebra on Ω is a σ -algebra.

Proof. Let $\mathcal{A}$ be a finite algebra on Ω , and we can write $\mathcal{A} = \{A_1, A_2, \dots, A_n\}$.

- Since $\mathcal{A}$ is an algebra, $\Omega \in \mathcal{A}$.
- Since $\mathcal{A}$ is an algebra, if $A_i \in \mathcal{A}$, then $A_i^c \in \mathcal{A}$.
- Since $\mathcal{A}$ is an algebra, if $A_i, A_j \in \mathcal{A}$, then $A_i \cup A_j \in \mathcal{A}$.

Let $\{B_i\}_{i \in I}$ be a countable family of subsets of Ω , where I is a countable index set, and each $B_i \in \mathcal{A}$. Since $\mathcal{A}$ is finite, the image of the function $f : I \rightarrow \mathcal{A}$ defined by $f(i) = B_i$ is finite. That is, there are only finitely many distinct sets in $\{B_i\}_{i \in I}$. Let these distinct sets be $C_1, C_2, \dots, C_m$, where $m \leq n$ and $C_k \in \mathcal{A}$ for all $k = 1, 2, \dots, m$.

The countable union $\bigcup_{i \in I} B_i$ can therefore be written as:

$$\bigcup_{i \in I} B_i = C_1 \cup C_2 \cup \dots \cup C_m.$$

Since $\mathcal{A}$ is an algebra, it is closed under finite unions. By induction:

- For $m = 1$, $C_1 \in \mathcal{A}$.
- Assume $C_1 \cup C_2 \cup \dots \cup C_k \in \mathcal{A}$ for some $m = k \geq 1$.
- Then, $(C_1 \cup C_2 \cup \dots \cup C_k) \cup C_{k+1} \in \mathcal{A}$ because $\mathcal{A}$ is closed under binary unions.

Thus, $C_1 \cup C_2 \cup \dots \cup C_m \in \mathcal{A}$, and we conclude:

$$\bigcup_{i \in I} B_i \in \mathcal{A}.$$

Therefore, $\mathcal{A}$ is a σ -algebra.

□